Artificial Intelligence Review (2021) 54:6245–6276
https://doi.org/10.1007/s10462-021-09977-z

Recent developments in computational color image denoising with PDEs to deep learning: a review

Nadeem Salamat[1] · Malik Muhammad Saad Missen[2] · V. B. Surya Prasath[3,4,5,6]

Published online: 5 March 2021
© The Author(s), under exclusive licence to Springer Nature B.V. part of Springer Nature 2021

Abstract
Image denoising methods are of fundamental importance in image processing and artificial intelligence systems. In this review, we analyze the traditional and state of the art mathematical models for computational color image denoising. These algorithms are divided into methods that are based on the partial differential equations, low rank, sparse representation and recent developments based on deep learning models. These algorithms also compared in terms of image quality measures. Our analysis and review of the computational color image denoising filters indicate that the convolutional neural networks from the deep learning domain obtain high quality restorations in terms of image quality despite the higher computational complexity.

Keywords Color image denoising · Partial differential equations · Deep learning · Low rank · Sparse representation · Convolutional neural networks · Computational imaging

✉ Nadeem Salamat
nadeem.salamat@kfueit.edu.pk

Malik Muhammad Saad Missen
saad.missen@iub.edu.pk

V. B. Surya Prasath
surya.iit@gmail.com; prasatsa@uc.edu

1 Department of Mathematics, Khawaja Fareed University of Engineering and Information Technology, Rahim Yar Khan 64200, Pakistan

2 Department of Computer Science and IT, The Islamia University of Bahawalpur, Bahawalpur 63100, Pakistan

3 Division of Biomedical Informatics, Cincinnati Children's Hospital Medical Center, Cincinnati, OH 45229, USA

4 Department of Pediatrics, University of Cincinnati College of Medicine, Cincinnati, OH 45257, USA

5 Department of Biomedical Informatics, College of Medicine, University of Cincinnati, Cincinnati, OH 45267, USA

6 Department of Electrical Engineering and Computer Science, University of Cincinnati, Cincinnati, OH 45221, USA

1 Introduction

The natural images are corrupted with the noise due to the thermal effects in image acquisition system, varying spatial resolution, moving objects in scene, online data transferee, compression and many other reasons. Clean and enhanced edges are the requirement of many post processing applications such as object detection, labeling, image segmentation and image interpretation. This gives rise to the image denoising models that are building blocks in devising robust artificial intelligence methods for visual data. The modern applications use the color and multispectral images directly, as a result, the methods which can denoise the color images are of fundamental importance. The grey image denoising methods are extended in two ways, use the grey image denoising method directly to color images via plane by plane or develop the method using the vectorial representation of images.

In this work, we consider computational denoising methods that are based on partial differential equations (PDEs) and artificial neural networks (ANNs). In literature, some surveys works are found in specific areas of color image denoising, such as application of vectorial PDEs (Tschumperle and Deriche 2007; Pal et al. 2015; Guichard et al. 2002), neighborhood filtering methods (Venetsanopoulos and Plataniotis 2000; Buades et al. 2005) and recent developments (Goyal et al. 2020). All these are the collection of work in the field of diffusion PDEs. No such survey work is found in other areas such as algorithms related to filtering in spatial or frequency domain and the newly emerging area of deep learning via ANN models. Following are the main objectives of the paper.

- This paper covers the methods for color image denoising using PDEs approaches, low rank, sparse representation and artificial neural network (ANN) based methods
- The objective of this survey is to analyze the efficiency of algorithms by Image Quality Measures (IQMs)
- The multi spectral and hyper spectral images are considered as the special case of RGB color images, so the methods which deal with multi spectral or hyper spectral images are also discussed

The rest of this survey work is structured as follows. The noise models and image quality measures (IQM) are discussed in Sect. 2 and the major computational color image denoising models are detailed in Sect. 3. Section 4 concludes the review.

2 Image noise models and image quality metrics

In this section, we discuss the basic image noise models and the image quality metrics, which are frequently used in reported models. This section is further divided into two subsections.

2.1 Different image noise models

Image and image sequences are corrupted the data during the acquisition, transmission and storing procedures due to various types of noise and edge diffusion and blurring. The complicated image structures and spatial organization of objects does not allow

to detect these distortions exactly. To detect these distortions and designing the image denoising methods, the noise models are important to understand. These noise models are listed in detail in Boncelet (2009). Let $y(\cdot)$ denotes the out put image and $x(\cdot)$ represents the original image and noise is represented as $q(\cdot)$ then noise models are classified as follows:

Additive Models: The commonly known additive noise model is described as

$$y(t) = x(t) + q(t). \tag{1}$$

In this forward model, noise is independent in the model from the output image. Examples are the thermal noise (Additive Gaussian noise), quantization noise and photographic noise. Quantization noise is produced during the quantization of random variables to the discrete variables. The noise with the uniform power, also known as Additive White Gaussian Noise (AWGN) for uniform Gaussian distribution function.
Multiplicative Models: The multiplicative noise model is described as the multiplication of a noise factor with the clean image.

$$y(t) = x(t) \cdot q(t) \tag{2}$$

This model represents noise when output image is noise dependent. Speckle Noise is an example of this noise. The multiplicative model can be transferred into additive and vice versa by applying the logarithm and exponential operators.
Mixed Noise (Neither independent nor dependent Noise There are some types of noise which do not fall into any category. The Salt and Pepper Noise which is produced during transmitting image. Poisson Noise are the examples of this type of noise. These are further explained as:

- Salt and Pepper Noise (Mukhopadhyay and Mandal 2014): The noise is called salt and pepper (SPN) or impulse noise when the dark regions have bright pixels and bright region have dark pixels. This type of noise may be caused by analog to digital converter errors, bit errors in transmission.
- Random Valued Noise Model: Noise in this model appears with a fixed length of m, by two fixed ranges. For example, if $m = 30$, noise will be equally likely to be any values in the range of either [0, 29] or [226, 255]. Similarly, if m is 256, then noise will equal likely be any values in the range of [0, 255]. Mathematically, it is defined as

$$f(n) = \begin{cases} \frac{p1}{2} & \text{if } 0 \le x \le m, \\ 1 - p & \text{for } x = y, \\ \frac{p2}{m} & \text{for } 255 - m < x < 255, \end{cases} \tag{3}$$

where p is the density of noise.

Initially, image denoising methods were developed for grey images and these methods focus on a particular noise type, such as Gaussian noise, additive noise, Additive White Gaussian Noise (AWGN), impulse and speckle noise. These methods are extended to deal the multichannel images in many ways, for example cross channel denoising, channel coupling and channel by channel denoising. These methods have provided important contributions in specific noise reduction from color images.

2.2 Metrics used for testing the efficiency of color image denoising models

The image quality is an image character through which image degradation, distortion or artifacts are measured with reference to the original image. The statistical measures are used to measure the objective image quality. Let X and Y are original and processed image and commonly used statistical measures are given in Table 1. Note that other metrics commonly used in grey scale image denoising can also be extended to handle color images by averaging across channels.

3 Recent image denoising models

In this section, color image denoising models are described in detail, and these models are further divided into the following three classes:

- Partial differential equations (PDEs) based image denoising approaches
- Low rank, and sparse representation models
- Methods based on deep neural networks

All these methods are summarized in the following subsections in terms of their main formulations, advantages, and we provide comparisons of their experiments based on different reported quality metrics.

3.1 Methods based on partial differential equations (PDEs)

The image regularization is considered as filters where variation in image pixels is minimized. The image regularization using anisotropic diffusion Partial Differential Equations (PDEs) is initiated by Perona and Malik (1990) for grey image denoising. The term anisotropic diffusion is used for image smoothing in parallel directions to boundary of an object and this process is stopped across the boundaries. For multichannel image denoising and regularization, these approaches are extended and color images are treated as multi-dimensional vectors where each image channel is considered a grey image and channel correlation is used. These methods are also extended for other color spaces, where the channels are uncorrelated. Tschumperle and Deriche (2007) reviewed the role of PDEs. For this review, we divide the PDEs based approaches into the following subcategories:

- Variational methods: (Energy functional is minimized)
- Divergence based diffusion PDEs ($g(|\ v\ |)$ is a stopping criteria)
- Oriented heat flow PDEs (heat equation is used with directional coordinates)
- Trace based diffusion PDEs (Tensor based diffusion)

In the following, we explain all these methods in detail, and review their corresponding results.

Table 1 The image quality metrics which are frequently used for results comparison in image denoising models for color images

Metric	Mathematical representation	Interpretation
MAE	$\frac{1}{MN}\sum_{i=1}^{m}\sum_{j=1}^{n}(x_{ij}-y_{ij})$	Lower MEA describes the better results and it is extended as mean of three channel for color images
MSE	$\frac{1}{MN}\sum_{i=1}^{m}\sum_{j=1}^{n}(x_{ij}-y_{ij})^2$	Lower MSE describes the better performance of algorithm. Mean of MSE is taken for multichannel images
PSNR	$10\log_{10}\frac{255^2}{MSE}$	Directly proportional to image quality and it is extended for color images as mean of three RGB channels
SSIM	$\frac{(\mu_X\mu_Y+C_1)(2\sigma_{XY}+C_2)}{\mu_X^2+\mu_Y^2+C_1)(\sigma_X+\sigma_Y+C_2)}$ where μ_X, μ_Y and σ are mean intensity values and standard deviation respectively. The σ_{XY} is the covariance of X, Y and C_1 C_2 are constants	This measure is extended to Multiscale Structural Similarity Index (MSSIM) in Wang et al. (2003) and Hassan and Bhagvati (2012). This captures the structural loss in images. Normalized higher value (close to 1) represents the better results
FCSS	$S(X_W;Y_W)=(SC)^{\alpha}(SS)^{\beta}(SL)^{\gamma}$ where C_{X_W} and C_{Y_W} are contrast (SC), structure (SS) and luminance (SL) measures computed patch wise for each channel. The parameters $\alpha,\beta,\gamma>0$ are used to adjust relative importance of three components	Combination of three measures defined in Grecova and Morillas (2016). The similarity measure will be high if all three similarities are high and it is directly proportional to the similarity between images. Let x_i and y_i are pixels and $\overline{X}_w$ and $\overline{Y}_w$ are mean color vectors in fuzzy metrics X_W and Y_W
FSIM	$\frac{\sum_{x\in\Omega}S_l(x)\cdot PC_m(x)}{\sum_{x\in\Omega}PC_m(x)}$ Here, measure Sl is computed as $S_l(x)=[S_{PC}(x)]^{\alpha}[S_G(x)]^{\beta}$ where α,β are adjustment parameters and Phase Congruency is computed for maximum Fourier coefficients. The local amplitude is defined as $X_{n,\theta_j}(x)=\sqrt{e_{n,\theta_j}(x)^2+O_{n,\theta_j}(x)^2}$ and local energy along orientation θ_j is $E_{\theta_j}=\sqrt{F_{\theta_j}(x)^2+H_{\theta_j}(x)^2}$ where $F_{\theta_j}(x)=\sum_n e_{n,\theta_j}(x)$ and $H_{\theta_j}(X)=\sum_n O_{n,\theta_j}(x)$. Then 2D PC is computed as: $PC(x)=\frac{\sum_j E_{\theta_j}(x)}{\sum_n\sum_j A_{n,\theta_j}(x)}$ where $PC\in[0,1]$. The similarity measure of PC of two images is defined as $S_{PC}(x)=\frac{2PC_1(x)\cdot PC_2(x)+T_1}{PC_1^2+PC_2^2+T_1}$ where T_1 is a positive constant and its value depends upon the dynamic range of PC. Similarly gradient feature of an image $G=\sqrt{G_x^2+G_y^2}$. The gradient features of two images are represented by $G_1(x)$ and $G_2(x)$, then gradient similarity measure is computed as $S_G(x)=\frac{2G_1(x)\cdot G_2(x)+T_2}{G_1^2+G_2^2+T_2}$ where T_2 is a positive constant and its value depends upon the Image Gradient Magnitude (GM) range	The Feature Similarity index Measure (FSIM) (Zhang et al. 2011) is computed in two stages. Two types of features PC and GM are used in this measure. Normalized higher value (close to 1) of FSIM stands for good resemblance of images measure

Table 1 (continued)

Metric	Mathematical representation	Interpretation
NCD	$\frac{\sum_{j=1}^{N}(\sqrt{(L_j-\hat{L}j)(a_j-\hat{a}_j)(b_j-\hat{b}_j)})}{\sum_{j=1}^{N}(\sqrt{(L_j^2+a_j^2+b_j^2)})}$ where $L_j, a_j, b_j, \hat{L}_j, \hat{a}_j, \hat{b}_j$ are coordinates of original and restored image pixels in *Lab* color space.	Normalized Color Difference (NCD) is equivalent to MSE in *Lab* color space (Melange et al. 2011). The lower NCD value indicates that image is similar to reference image
IFS	$(IFS_{lum}^{\alpha} IFS_{fea}^{\beta})^{\frac{1}{\alpha+\beta}}$ where α, β are balancing parameters between Luminance and feature components with positive values. Feature similarity defined as $\frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}\left(\frac{2X_{ij}Y_{ij}+c}{(X_{ij})^2+(Y_{ij})^2+c}\right)$ and its luminance part as $\frac{\sum_i(m_i^{ref}-\mu(m^{ref}).(m_i^{dis}-\mu(m^{dis})+c_m}{\sqrt{\sum_i(m_i^{ref}-\mu(m^{ref})^2\cdot(m_i^{dis}-\mu(m^{dis})^2+c_m}}$	Independent Feature Similarity (IFS) Index consist of two parts. The features are computed by matrix multiplication $F^{fea} = WA, F^{dis} = WB$ where F describe the two independent feature sets extracted from W
SAM	$\arccos(\frac{\langle A,B\rangle}{\lvert A\rvert\lvert B\rvert})$	The SAM is used to measure the quality of fused images. The lower value presents better results and shows that image is similar to reference image. It computes the spectral angle between the pixel, vector of the reference image and fused image. This can be represented in both degrees and radians. A lower value of SAM indicate the spectral distortion is decreased between images
ERGAS	$100\frac{dh}{dl}\left[\frac{1}{n}\sum_{i=1}^{n}(\frac{RMSE^2}{mean^2})\right]^{\frac{1}{2}}$ where $RMSE^2 = MSE$ represents the root mean square error.	Image quality is inversely proportional to this measure, i.e., image quality is better for lower values of this measure. This measure is employed to measure the image quality in fused images and it shows that how much the distortion in the fused images. The lower value of ERGAS shows that the fused image is similar to the reference image (Wald 2002)

3.1.1 Variational methods

In the variational methods, the image regularization problem is also considered as global minimization procedure. In this method, image I is recovered from noisy image observation I_{Noisy} by minimizing the cost functional. These sort of PDEs minimize the adapted variational functional and gradually remove noise, while preserving the contours in image regions. These functionals can be described as,

$$\min_{I:\,\Omega\to R^2} = \int_{\Omega} \phi(\mid \nabla I \mid)\, d\Omega,$$

where $\phi: R \to R$ is an increasing function which perform the regularization and minimization is performed via corresponding diffusion PDE obtained from Euler-Lagrange equation of $E(I)$, such that,

$$I_0 = I_{noisy},$$
$$\frac{\partial I}{\partial t} = div\left(\frac{\phi'(\nabla I)}{\mid\mid \nabla I \mid\mid}\right).$$

The choice of function ϕ lead to a separate regularization methods. Minimization of fidelity term in L^2 norm and some related results are provided in Fuchs et al. (2018), Convex regularization of multi-channel images based on variants of the TV-model with diffusion coefficients are selected based on certain criteria (Rafsanjani et al. 2017).

3.1.2 Contribution summaries

Moreno et al. (2016a) introduced a weighted TV regularization with coupling channels gradients. The channel u_k and u_l are coupled using functional $J(u_k, u_l) = \int_{\Omega} \mid \nabla u_k - \nabla u_l \mid^2 dx$. This is extendable for multi channels as $\sum_{k=1}^{N} \sum_{l=1}^{N} J(u_k, u_l) = \int_{\Omega} \sum_{k=1}^{N} \sum_{l=1}^{N} \mid \nabla u_k - \nabla u_l \mid^2 dx$. The coupling VTV of u is introduced as,

$$VTV(u) := \sum_{k=1}^{N} \int_{\Omega} \mid Du_k \mid^2 dx + \alpha \int_{\Omega} \sum_{k=1}^{N} \sum_{l=1}^{N} \mid \nabla u_k - \nabla u_l \mid^2 dx, \tag{4}$$

where $\alpha \geq 0$ for cross-coupling terms. Another version of TV with L^1 fidelity term for multichannel images,

$$E^1_{VTV}(u) = \sum_{k=1}^{N} \left(\int_{\Omega} g_k(x) \mid \nabla u_k \mid dx + \lambda \int_{\Omega} \mid h_k - f_k \mid dx \right) + \alpha \int_{\Omega} \sum_{k=1}^{N} \sum_{l=1}^{N} \mid \nabla u_k - \nabla u_l \mid^2 dx, \tag{5}$$

where regularization function $g_k(x) = (1 + \mid \nabla f_k(x) \mid^2)^{-1}$ is implemented and this functional is minimized.

Chan et al. (2001a) introduced the TV model for brightness and chromaticity channels. The vector valued TV and color TV models are proposed by Blomgren and Chan (1998). The brightness and chromaticity is denoised by grey scale and non-flat TV models separately. The classical TV model can be written as the following minimization problem,

$$\min_u \int_\Omega \mid \nabla u \mid dxdy + \frac{\lambda}{2}\int_\Omega (u-u_0)^2 dxdy. \tag{6}$$

This is minimized using the Euler–Lagrange method and TV model for chromaticity developed in Chan and Shen (2001) is implemented. However, it is well known that digital TV filters create the staircasing artifacts in flat regions which can be detrimental in further downstream image processing operations such as image segmentation and pattern recognition. To avoid this various improvements can be considered including higher order PDE filters or adaptive weighting functions along with TV regularization.

Chan et al. (2001b) also introduced a graphical digital TV filter for color image denoising. The signal energy u is defined as $\mid \nabla_\alpha u \mid := \sqrt{\sum_{\beta\sim\alpha} u_\beta - u_\alpha}$. This energy with a local variation becomes $\mid \nabla u_\alpha \mid_\alpha = \sqrt{\mid \nabla u_\alpha \mid^2 + \alpha^2}$. The directional derivatives along graph edge $\alpha \sim \beta$ are defined as $\frac{\partial u}{\partial e}\mid_\alpha = u_\beta - u_\alpha$, where (a) and λ are regularization and fitting model parameters respectively. The mapping $F^{\lambda,a} : u \to v$ where u and v are input and output signal. Its value at node α is $v_\alpha = F_\alpha(u) = \sum_{\beta\tilde{\alpha}} h_{\alpha\beta}(u)u_\beta + h_{\alpha\alpha}(u)u_\alpha^0$ and the filter coefficients are defined as $h_{\alpha\beta} = \frac{w_{\alpha\gamma}}{\lambda+\sum_{\gamma\tilde{\alpha}} w_{\alpha\gamma}}$ and $h_{\alpha\alpha} = \frac{\lambda}{\lambda+\sum_{\gamma\tilde{\alpha}} w_{\alpha\gamma}}$ where $w_{\alpha\beta} = \frac{1}{|\nabla_\alpha u|_\alpha} + \frac{1}{|\nabla_\beta u|_\alpha}$. Therefore, h is a low pass filter. This scalar TV filter is extended for color images as $I : \Omega \to R^3$ where I_α is a color pixel and $\mid \nabla_\alpha I \mid = (\sum_{\beta\tilde{\alpha}} \mid I_\beta - I_\alpha \mid^2)^{\frac{1}{2}}$, this filter is called the vectorial digital TV filter. This filter is also derived in spherical components.

The scalar TV model can also be cost as follows,

$$TV(u) = \left\{ \int_\Omega u div(\varphi) dx,\ \varphi \in C_c^1(\Omega, R^2), \mid\mid \varphi \mid\mid_\infty \leq 1 \right\}, \tag{7}$$

and this is extended for color images applicable in opponent color spaces by Astrom and Schnorr (2017). The tensor is decomposed using the Eigenvalue decomposition method and signal energy is represented by $\min_u\{E(u) = \frac{1}{s} \mid\mid Ku - g \mid\mid_s^s + \sum_{m=1}^{M}(\frac{a_m}{p_m} \mid\mid D^m(u) \mid\mid + \frac{b_m}{q_m} \mid\mid D^m Cu \mid\mid)$ where k is the identity matrix and DCu is pair wise mixture of channel gradients. This energy is minimized and VTV in double opponent color space,

$$VTV(u) = \int_\Omega \mid\mid \nabla Cu \mid\mid := \sup_{\mid\mid p \mid\mid_p \leq 1} \int_\Omega \langle c_1 u, div(p) \rangle dx. \tag{8}$$

The regularization term is convex and intensity invariant. It is optimized using the Half Quadrature Analysis (HQA) and Split Bregman algorithms (Prasath and Moreno 2018).

Bar et al. (2005) introduced the functional

$$F = \int_\Omega \sum_\alpha \phi(h * u^\alpha - z^\alpha) dA + J(u), \tag{9}$$

for color image denoising, where $\phi(\cdot)$ and $J(u)$ are potential and regularization functions respectively. The fidelity terms $\phi(h * u^\alpha - z^\alpha) = (h * u^\alpha - z^\alpha)^2$ and $\phi(h * u^\alpha - z^\alpha) = \sqrt{(h * u^\alpha - z^\alpha) + \eta}$ are suitable for Gaussian and impulse noise. The authors used the color image deblurring functionals with L^1 fidelity term. The regularization terms $J^{BEL}, J^{MS}, J^{MSTV}$ are used channelwise and minimization is performed. Following the Beltrami framework introduced by Sochen et al. (1998), a color image $(u_r;u_g;u_b)$ is represented as a $2D$ surface in R^5-space and the Beltrami functional is given as, $J^{BEL} = \alpha \int_\Omega \sqrt{Det(G)} dA$ is rewritten as

$$\int_{\Omega} \sqrt{1+\gamma^2 \sum_{\sigma}(u^{\sigma})^2 + \frac{\gamma^4}{2} \sum_{\sigma_1,\sigma_2}(\mid \nabla u^{\sigma_1} \times \nabla u^{\sigma_2} \mid^2)}\, dA, \tag{10}$$

and channel alignment is obtained by cross product. The classical Mumford and Shah (1989) functional $J^{MS}(u,k) = \beta \int_{\Omega \setminus K} \mid \nabla u \mid^2 dA + \alpha \int_k dH^1$ and its extension for color images is approximated by regular functionals in Ambrosio and Tortorelli (1990) by,

$$J_{\epsilon}^{MS}(u,v) = \beta \int_{\Omega \setminus K} v^2 \mid \nabla u \mid dA + \alpha \int_{\Omega} \left(\epsilon \mid \nabla v \mid^2 + \frac{v^2-1}{4\epsilon} \right) dA. \tag{11}$$

A modified version of Shah (1996) functional,

$$J_{\epsilon}^{MSTV}(u,v) = \beta \int_{\Omega \setminus K} v^2 \mid \nabla u \mid dA + \alpha \int_{\Omega} \left(\epsilon \mid \nabla v \mid^2 + \frac{v^2-1}{4\epsilon} \right) dA, \tag{12}$$

is generalized for color images.

The gradient functional,

$$F_{\epsilon,p}(u) = \int_{\Omega} \mid \nabla^{\epsilon} u \mid^p dx + \lambda \mid\mid u_0 - u \mid\mid^2, \tag{13}$$

is minimized in the hybrid model proposed by Kim (2006), where u_0, λ and ϵ are observed image, Langrange multiplier and regularization parameter respectively. In this model, signal energy is defined as $\mid \nabla^{\epsilon} u \mid = (u_x^2 + u_y^2 + \epsilon^2)^{\frac{1}{2}}$. The model becomes total variation (TV) model for For $p = 1, \epsilon = 0$. This minimization problem is solved by Euler-Lagrange equation,

$$u_t = \nabla \cdot \left(\frac{\nabla u}{\mid \nabla^{\epsilon u} \mid^{2-p}} \right) + \beta(u_0 - u).$$

The minimization is performed by decent directions and $B = \frac{2\lambda}{p}$ is called the constraint parameter. The $p = 1 - w$ is a non convex parameter for edge preserving and β is a diffusion controlling parameter. The Enhanced Total Variation Model (ETVM) defined as

$$u_t = \nabla \cdot \left(\frac{\nabla u}{\mid \nabla^{\epsilon u} \mid^{1+w}} \right) + \beta(u_0 - u) \tag{14}$$

The target image is blurred for smaller β, this deficiency is improved by applying Motion by Mean Curvature (MMC) model, represented as

$$u_t = \nabla \cdot \frac{\nabla u}{\mid\mid \nabla u \mid\mid} \tag{15}$$

The minimum-biased difference is used as diffusion controlling parameter and $\mid \nabla u \mid$ is approximated by smallest two differences. Finally both algorithms, ETVM and MMC [Eqs. (14), (15)], are combined as,

$$u_t - \mid \nabla u \mid^{\alpha} \cdot \nabla \left(\frac{\nabla u}{\mid\mid \nabla^{\epsilon} u \mid\mid^{1+w}} \right) = \beta(u_0 - u) \tag{16}$$

where $\alpha, \beta, w > 0$. This model is extended for color images (Kim 2006) and brightness of color image is directed denoised using this method and angle domain method is used to denoise chromaticity.

3.1.3 Divergence based diffusion PDEs

In this sort of PDEs application to image denoising, the regularization is achieved by divergence expression. The idea is to replace the function $\frac{\phi'(\nabla I)}{||\nabla I||}$ by the divergence of a scalar valued PDEs using the appropriate image features,

$$\frac{\partial I}{\partial t} = div(g(\mid \nabla I * G_\alpha \mid) \mid \nabla I \mid),$$

where $*, G_\alpha$ stands for convolution and Gaussian mean noise. In these methods, it is supposed that the pixels act as temperatures which diffuse with respect to some physical laws. It is parameterized by a field $D\colon \Omega \to P(2)$, a diffusion tensor, $\frac{\partial I}{\partial t} = Div(D\nabla I_i)$ where D is a diffusion tensor and $D = \frac{\partial \psi}{\theta_+ \theta_+^T} + \frac{\partial \psi}{\theta_- \theta_-^T}$ where tensor $D = \lambda_1 uu^T + \lambda_2 vv^T$.

Wang et al. (2017b) extended the combination of MAD and energy functional

$$E(u_i) = \int_\Omega \left(\frac{\lambda}{2}(u_i - f_i)^2 + \psi(\mid \nabla u_i \mid^2) \right) dx \tag{17}$$

to the functional

$$E(u_i) = \int_\Omega \left(\frac{\lambda}{2}(u_i - f_i)^2 + \psi\left(\sum_{b=1}^{B} \mid \nabla u_i \mid^2 \right) \right) dx \tag{18}$$

where u_i and f_i are the ith bands of the noise-free and degraded multispectral images, respectively. The minimizer is achieved using Euler–Lagrange method and inter band correlation is achieved using gradient sum. The vectorial MAD is used for spectral adaptive mechanism and spatial adaptive weighted function $f(\cdot)$ is used to initiate the diffusion process at pixel level in MAD model. The SSAWBF-MAD is represented by

$$\frac{\partial u_i}{\partial t} = \phi(\mid \nabla u_\sigma \mid^2) div(g(\mid \nabla u_\sigma \mid^2) \nabla u_i) - \varphi(\mid \nabla u_\sigma \mid^2)(u_i - v_i) \tag{19}$$

where v_i and u_i are obtained and initial image. The two terms are balanced using sigmoid function $\varphi(\mid \nabla u_\sigma \mid^2) = \beta \frac{1 - e^{-\gamma |\nabla u_\sigma|^2}}{1 + e^{-\gamma |\nabla u_\sigma|^2}}$ where β moderates the fidelity term. This model assure a spectral-spatial adaptive and selective smoothing that removes noise without producing undesirable structures. The weighted function was adapted from Prasath and Vorotnikov (2014) and it balances noise reduction and edge preservation during the iterative denoising process.

A diffusion PDE based algorithm for multispectral image denoising is proposed by Prasath and Singh (2010). The N-dimensional version of the Perona and Malik diffusion PDE (Perona and Malik 1990) is $\frac{\partial u^i}{\partial t} = div(g(\mid \nabla u^i \mid \nabla u^i),\ i = 1, 2, \ldots, N$ where $g \in [0, \infty]$ is a diffusion coefficient and u_0^i, t_2 are initial image and time variables respectively. The actual choices for g are $g(s) = e^{-\frac{s^2}{k}}$ and $g(s) = (1 + \frac{s^2}{k})^2$ where K is a constant. The channel coupling term is added to the n-dimensional anisotropic diffusion PDEs, defined as $\frac{\partial u^i}{\partial t} = div(g(\mid \nabla u^i \mid \nabla u^i) + \sum_{j=1}^{N}(w^i \triangle u^i - w^j \triangle u^j)$ for all $i = 1, 2 \ldots, N$ channels. The second term in PDE represents the inter channel coupling and diffusion balancing relation

and adaptive weights are chosen as $w^i = \nabla_\rho u_0^i = G_\rho * \nabla u_0^i$ where $*$ is convolution and G, ρ are the Gaussian kernel and edge smoothing parameter respectively. Other options are available for selecting the adaptive weights, for example, using TV denoised channels before taking the smoothed gradients (Prasath 2011).

3.1.4 Oriented heat flow PDEs

Two mutually orthogonal components of a smoothing vector is computed at each point and heat flow is controlled in the direction of u_1, u_2. The relevant PDE is written as the sum of two heat flows as $\frac{\partial I}{\partial t} = c_1 I_{u_1 u_1} + c_2 I_{u_2 u_2}$ where u_1, u_2 are orthogonal vectors and c_1, c_2 are constants.

3.1.5 Trace based diffusion PDEs

The trace based diffusion PDE can be considered as a generalization of oriented Laplacian. It is the rewrite up of oriented heat flow equation by the trace operator,

$$\frac{\partial I}{\partial t} = Trace(TH_i),$$

where H is the hessian matrix at each point and $T = c_1 I_{u_1 u_1} + c_2 I_{u_2 u_2}$. In this case, the regularization problem is converted into the design of unique tensor T. Tschumperle (2006) used the traced based PDEs for multiple color image applications such as image denoising, inpainting and image resizing.

Deblurring and denoising color images by PDEs is suggested by Barash (2001). In this algorithm, noise with point spread function is considered and model is represented by $f(x;y) = (d * u)(x;y) + n(x;y)$, where $u(x;y), d, *, n$ describe the original image, blur PSF, convolution and additive noise respectively. Objective is to reconstruct u from f. For this purpose, the image norm that represents the smoothness, denoted by $\psi(\mid u \mid)^2$ is minimized. $\min_{u(x,y)} \int_\Omega \psi(\mid \nabla u(x,y) \mid)^2$ with constraints $\int_\Omega (d * u - f) 2dxdy = \sigma^2$ where σ is the standard deviation of noise. The minimization problem is reformulated as the minimization of energy function. $\min_{u(x,y)} \int_\Omega \psi(\mid \nabla u(x,y) \mid)^2 \alpha (d * u - f) 2dxdy = \sigma^2$, where α is a Lagrange-multiplier can be determined by calculus of variations. The choice of an appropriate ψ is important because the results are ψ' dependent. In this algorithm, $\lambda \sqrt{1 + (\frac{|\nabla u|}{\lambda})^2}$ is used and equation is solved numerically. Ma et al. (2007) applied a fuzzy vector partition filtering technique for color image restoration. All these methods works well for Gaussian noise and experiments are performed for Gaussian noise with multiple noise ratios.

3.1.6 Results comparison

The results for the PDEs application to color images are compared in the Table 2. In this table, the MPSNR, MMSE and MSSIM are means of channelwise PSNR, MSE and SIMM measures respectively. As can be seen, all the aforementioned PDE based methods works well for different Gaussian noise levels. Note that in Tschumperle (2006), possible applications in color image processing are discussed however no comparison results are given.

Table 2 PDE based color image denoising methods

Method	PDEs type	Images, noise ratio(%)	MPSNR	MMSE	MSSIM
FAS (Tschumperle 2006)	Trace B	Baboon, Lena, 1.5	–	–	–
GCIRD (Astrom and Schnorr 2017)		Multiple p values, 80	PSNR $(p1, p2) = (23.4 \pm 1.8, 23.2 \pm 1.9)$		SIMM = $(0.96 \pm 0.09, 0.69 \pm 0.09)$, CIEDE = $(6.91 \pm 2.28, 7.14 \pm 2.32)$
SSAWBF-MAD (Wang et al. 2017b)	Div	Lena, Parrots, House, 30	28.42, 28.78, 29.66	–	0.8108, 0.8464, 0.7869
VTV (Moreno et al. 2016b)	TV	Berkeley data set, 20 and $\lambda = 2^{-4}$	PSNR = 28.1685	MSE = 126.6188	MSSIM = 0.7589
MSADS (Prasath and Singh 2010)	Div	Satellite images, 30	PSNR = 31.7, 30.75	–	–
ABO (Kim 2006)	TV	Vegetables, 30	–	–	–
VTV (Chan et al. 2001a)	TV	Vegetables, street signs	–	–	–
DTV (Chan et al. 2001b)	TV	Vegetables, street signs, 30	–		
CIDIN (Bar et al. 2005)	TV	Lena, 30	PSNR= 21.93, 23.28, 23.2	–	–
DCI (Barash 2001)	Div	Packard, circles	–	–	–

3.2 Sparse representation

The sparse representation has been utilized in many areas of computer vision and image processing, in which image signals can be represented as the linear combination of basis elements from dictionary. To implement such methods, the whole image X is divided into patches, called y. These patches are represented as a column vector and can be used in tensor or singular decomposition based approaches for obtaining state of the art denoising results compared to the regularization or PDE models. In color image denoising methods, two properties are important, inter band correlation and the orthogonal of color channels. Based on these two properties, all these methods are divided into two subcategories:

- Tensor based methods: In these methods, the image matrix is decomposed into tensors. These methods preserve well the inter band correlation and does not necessarily obey the orthogonal property.
- Singular decomposition methods: In these methods, the singular decomposition property of matrixes is employed. These methods preserve the orthogonal property of channels

Methods of both these paradigms with salient details are explained in the following subsections along with comparisons based on their experimental results and quality metrics.

3.2.1 Prior image denoising model

The search of similar patches in images is adopted in Gaussian prior image denoising models (Hu et al. 2019). The image noise model is described as $y = x + n$ where x, y are the clean and noisy images and n is a Gaussian noise. In this model, each patch have the Gaussian distribution and MAP estimator with Gaussian parameters is used to obtain the denoised image,

$$X_i = \underset{x_i}{argmin} \frac{1}{2\sigma^2} \mid\mid y_i - X_i \mid\mid_2^2 + \frac{1}{2}(x_i - \mu_i)^T \sum_i^{-1} (X_i - \mu_i).$$

The result X_i is a balance between the fidelity term and the prior term and it is the problem of convex optimization.

In image prior methods, nonlocal similar patches are assumed to be a Gaussian distribution, and these samples are used to train the parameters. The mean is computed using the maximum likelihood. All the patches in noisy data, may have different Gaussian distribution and this problem is handled by using the weighted mean.

The covariance of the model is obtained by adding the semi definite terms and it is obtained by eigenvalue decomposition $y_i = USU^T$ and equation is solved. This method does not work well for high level noise. The accuracy in parameters is obtained by using iterative methods. For smooth images have small variance and these are used for patch variance and eigenvalue decomposition $\Sigma_i = \sum s_i u_i u_i^T$. The performance of the algorithm is increased by using iterative regularization.

3.2.2 Low rank minimization

A tensor is the multidimensional data represented as a vector mapping over the vector spaces and tensor representation of image data helps to preserve some image structural

details. The simple inter and intra channel computation sometime causes information loss in image denoising steps. Following are the contribution summaries.

An adaptive frame based color image denoising is proposed by Shen et al. (2016), in which color images are divided into patches. A block Y of a color image, is a $p \times p \times 3$ subimage of approximated image Z, known as reference patch. The Euclidian distance based similarity between the reference and other patches are determined and $k-1$ similar patches are marked. A fourth order tensor $Y = R^{p \times p \times 3 \times K}$ is formed with $Y(:,:,:,k) = Y_k, k = 1, \ldots, K$. Every tensor is decomposed using the higher order singular decomposition using the hard threshold. The denoised image is reconstructed using the inverse transformation and weighted averages. For complete reconstruction, every pixel must be part of a reference image patch. The residual image is updated with a threshold. This updating procedure is continued till the required criteria is met.

Moreno et al. (2011) used tensor voting methodology for color image denoising. The canonical voting fields (Medioni et al. 2000) were designed to propagate the encoded information where noise is modeled using the plausible hypothesis and normal vectors represents the smoothness in surfaces. The voting is defined based on the color difference, uniformity and edge preserving requirement of the color image. The neighborhood and voting tensor for each pixel is determined using the positional and color information of a pixel. The color perception system is developed using CILAB (Wyszecki and Stiles 2000) image representation model and every channel is normalized. The tensors are represented by symmetric positive semi definite matrices and 2D tensors are used for encoding of a color pixel. The color is represented by the largest Eigen value and phase of the Eigen vector. The smooth and edge regions are represented by stick and ball type tensors and these tensors vote for every possible location in the image. The voting process is initiated by normalized saliency $s^1 = 1,\ s^2 = 0$. This algorithm requires less memory as compared to its counter parts. This tensor is propagated by convolution like process in each channel independently. The propagation function considers the tensor information and neighborhood relationship between pixels, stick (S_c) and a ball (B_c) propagation function are employed. The total votes are represented by

$$TV_c(p) = \sum_{q \in nhbd(p)} (S_c(P,q) + B_c(P,Q))$$

where p, c represents pixel and color channel respectively. The eigenvalues and eigenvectors are used for local color perception information, perceptual color difference is computed using formula developed in Luo et al. (2001).

The intrinsic tensor sparsity for multispectral image denoising is used by Xie et al. (2016). The original image z is separated form noisy image Y, which is divided into a collection of patches $\Omega = \{y_i \in R^d\}_{i=1}^N$. A set of patches similar to patch y_i is extracted and the patch matrix is recovered by $X_i = argmin_X S(X) + \frac{\lambda}{2} \mid\mid Y_i - X \mid\mid_F^2$ where $S(X)$, λ is a sparsity measure on the true matrix and controlling parameter respectively. The pixel wise combination of x_i is used to recover image z. The noisy image $Y \in R^{d_H \times d_W \times d_S}$ is represented by a 3 order tensor, where the d_H, d_W and d_S as the spatial height, width and spectral band number of an Multispectral Images (MSI). A MSI is represented by a group of 2D Full Band Patch (FBPs) $\{P_{ij}\}, 1 \le i \le d_H - d_h, 1 \le j \le d_W - d_w \subset R^{d_h d_w \times d_S}$ with ordered lexicographically as a column vector and tensors of order three, are constituted by using the non local self similarity (NSS) with FBPs collection of MSI. The true non local similarity FBPs x_i from noisy y_i is estimated by optimization of $X_i = argmin_X S(X) + \frac{\lambda}{2} \mid\mid Y_i - X \mid\mid_F^2$. The second order sparsity measure is extended to

higher order by weighted sum of ranks along all tensor modes. The tensor is decomposed using both, Tucker and CP decomposition. The intrinsic tensor sparsity (ITS) is defined as $S(X) = t \mid\mid S \mid\mid_0 + (1-t)\prod_{i=1}^{N} rank(X_i)$, where S is the core tensor of X obtained from the Tucker decomposition and $0 < t < 1$ is a balancing parameter of two terms. It is a long sum and hard to compute. So the above functional is changed and optimization is performed to new functional $Min_x P_{ls}(S) + \lambda \prod_{i=1}^{3} P_{ls}^{*} + \frac{\beta}{2} \mid\mid Y_i - X \mid\mid_F^2$ where $\lambda = \frac{1-t}{t}, \beta = \lambda t,\ P_{ls}(A) = \sum_{i_1,i_2,i_3} log(\mid a_{i_1,i_2,i_3} \mid + e)$, and $P_{ls}^{*}(A) = \sum_j log(\delta_j(A) + e)$, which are the log-sum forms of vector and matrix sparsities, respectively and e is a small positive number. The term $\delta_j(A)$ defines the jth singular of A.

A hyperspectral image denoising model with additive Gaussian noise is considered by Kong et al. (2019). In this model, the LR prior, and matrix representation of image denoising is taken as described in Gu et al. (2014).

$$\underset{X}{Arg\,min} = \frac{1}{2} \mid\mid Y - X \mid\mid + \mu Rank(X),$$

where μ is the tradeoff parameter. Similarly, the denoising problem of HSI in tensor format can be described as

$$\underset{X}{Arg\,min} = \frac{1}{2} \mid\mid Y - X \mid\mid + \mu Rank(X).$$

In the low-rank approximation problem, the nuclear norm is usually introduced as the surrogate functional of LR constraint. The HSI is treated as a 3-order tensor, so N = 3, and parameters α_1, α_2 should be equal and they should be smaller than α_3 when the denoising result is good. We have conducted various experiments and have empirically chosen $\alpha_1 = \alpha_2 = 0.2, \alpha_3 = 0.6$. The same object have the similar reflectance everywhere, so the object will have the same spectral measures in HSI representation. The image is decomposed into patches. The nonlocal similarity index among patches and intra channel global correlation information are usually used as prior knowledge in denoising methods. The inter band noise is removed globally and distortion noise is removed locally, using method proposed in LeCun et al. (2015), which uses nonlocal self-similarity (NSS) to characterize HSI. The Frobenius norm and Euclidean distances are used for computing the similarity. The original image is decomposed into overlapping patches. Then KNN algorithm is used for searching the similar patches. All these patches are grouped into a fourth order tensor, As the clusters have the similar structure, so these approximated using the LR tensor. The tensors are decomposed using the Tuker decomposition method. Both the nonlocal and global lowrankness is implemented and the regularization is performed using the unconstraint optimization method.

Color image denoising by monogenic signals is proposed in Gai (2019). The monogenic signal are the rotation invariant extension of analytical signals. The colored image in RGB space is represented and its gradient is determined and colored structure tensor is define as the sum of three channel structures. The colored monogenic signal is defined by four parameters and colored Riesz features are computed as a normal computation from Eigen values and Eigen vectors of the new tensor. Monogenic matrix singular value decomposition (MMSVD) is performed by using the normal SVD to the complex valued matrix. Thus a new sparse representation is proposed. The objective function x is used to collect the non zero signals in the colored image. As a result, the colored image is represented as the linear sum of three elements from the dictionary and the colored image is represented as

$$[X_r, X_g, X - b] = [D_r, D_g, D_b]\hat{m},$$

where $\hat{m}$ is the colored coefficient matrix. Let x_0 be the colored image in monogenic signal and white Gaussian is represented as G, then noisy observation $Y = X_0 + G$. Then the denoising problem is formulated as the minimization of the functional

$$[\tilde{\hat{B}}_{i,j}, \hat{D}, \hat{X}_0] = \min_{\hat{B_{i,j}}, \hat{D}, \hat{X}_0} \left\{ \lambda \mid\mid X_0 - Y \mid\mid_2^2 + \sum_{i,j} \mu_{i,j} \mid\mid B \mid\mid_0 + \sum_{i,j} \mid\mid \hat{D}\hat{B} - S_{i,j}\hat{X}_0 \mid\mid_2^2 \right\},$$

where the second and third terms are represented using the sparsity and indexes represents the position of a patch. The $S_{i,j}$ represents a square operator. The experiments are performed on many images.

Kong and Yang (2019), introduced the collaborative filtering method for multi spectral image denoising. In this method, image is divided into blocks(patches) and image is decomposed into similar patches. In CBM3D image denoising method, the group of similar image patches p is represented by G and fourth order tensor decomposition method is used (Kolda and Bader 2009). The patch level basis via block diagonal decomposition are developed and used as image denoising and tensor vector is written as $C = (G \times_3 U^T_{color}) \times_1 U^T_{row} \times_2 U^T_{column} \times_4 U^T_{group}$, where C is the core tensor. The $G \times_3 U^T_{color}$ color is equivalent to applying U_{color} to each patch of G via $G \times_3 U^T_{color} = P_i \times_3 U^T_{color}; i = 1;2; \ldots ;K$ where K and Pi represent number of similar patches and ith patch of G, respectively. In this method, the frontal slices of P_i are combined in matrices at diagonal position which is transformed into F diagonal. This develops a trade off between slice and patch relationship. The fourier transformation of initial two slices are computed and this is extendable for multispectral images. The $D \in R^{n \times k}$ sized sparse dictionary is prepared using the every element of patch as $a = arg_a \min \mid\mid a \mid\mid_0$.

3.2.3 Sparse representation

Elad and Aharon (2006a, b), proposed the image denoising via learned sparse dictionaries. In this model, the image patches are represented as column vectors and patches are denoised using MAP estimator by solving $\hat{a} = arg_a min \mid\mid a \mid\mid_0$ subject to $\mid\mid Da - y \mid\mid_2^2$ where $\mid\mid \cdot \mid\mid_0$ represents the non zero entries (Moreno et al. 2016b). This model is more precise for bounded error and denoised image is represented as $X = D\hat{a}$. The constraints is converted into penalty function and problems becomes the particular choice of penalty function μ. This is solved by matching and basis pursuit algorithms such as orthonormal matching pursuit (OMP). The image X is expressible as union of overlapping patches and every patch is the part of (ϵ, L, D) Sparseland model. For denoising, the MAP estimator is replaced by

$$\{a_{i,j}, X\} = \underset{a_{i,j}, x}{argamin} \lambda \mid\mid X - Y \mid\mid_2^2 + \sum_{i,j} \mu_{i,j} \mid\mid a_{i,j} \mid\mid_0 + \sum_{i,j} \mid\mid Da_{i,j} - R_{i,j}X \mid\mid_0 .$$

This is sum of proximity measure in Y, X and image prior with assurance of sparsity. The $n \times N$ matrix $R_{i,j}$ extracts the (ij) block from image and dictionary D is trained by,

$$\{\hat{D}, \hat{a_{i,j}}, \hat{X}\} = \underset{a_{i,j}, x}{argamin} \lambda \mid\mid X - Y \mid\mid_2^2 + \sum_{i,j} \mu_{i,j} \mid\mid a_{i,j} \mid\mid_0 + \sum_{i,j} \mid\mid Da_{i,j} - R_{i,j}X \mid\mid_0 .$$

The singular value decomposition (SVD) operation is performed on residual data matrices and dictionary is updated using columnwise K-SVD algorithm.

$$\hat{a_{i,j}} = \underset{a_{i,j}}{arg\,min}\mu_{i,j} \mid\mid a_{i,j} \mid\mid_0 + \mid\mid Da_{i,j} - R_{i,j}X \mid\mid_0 .$$

This is computed only on image patches and atoms along the update, represents the change in coefficients along with their sparsity structure. When all $\hat{a}_{ij}$ are updated, then these elements are fixed and $\hat{X}$ are updated. For this, the following equation

$$\hat{X} = \underset{x}{arg\,min}\lambda \mid\mid X - Y \mid\mid_2^2 + \sum_{i,j} \mid\mid Da_{i,j} - R_{i,j}X \mid\mid_0,$$

is solved. At reconstruction stage, the pixels can be represented as a linear sum of atomic dictionary elements. This algorithm is used for color image denoising via plane by plane and for colored images, we have $y_c = D_c a$ where D_c is the dictionary with k items.

Mairal et al. (2008) reviewed the method proposed by Elad and Aharon (2006a, b) and proposed the concatenation of color spaces, called concatenation model. In this method, a joint dictionary $[y_r^T;y_g^T;y_b^T]^T = [D_r^T;D_g^T;D_b^T]^T\alpha$ is developed. The dictionaries are prepared using 256 atoms with patch size $8 \times 8 \times 3$ and use K-SVD algorithm on (3-D) image patches. The produced color artifacts are removed by defining the new inner product. Let y and x are two patches written as column vectors $(R, G, T)^T$ then $\prec y, x \succ = y^T x + \frac{\gamma}{n} y^T K^T K x = y^T (I + \frac{\gamma}{n} K^T K)x$ where,

$$j = \begin{pmatrix} J_n & 0 & 0 \\ 0 & j_n & 0 \\ 0 & 0 & j_n \end{pmatrix}$$

Here, j_n is a $n \times n$ matrix, and γ is a new parameter. The average colors are used for atoms and change in metric in OMP is straightforward. A simple extension is replace $I + \frac{\gamma}{n}K$ with$(I + \frac{\gamma}{n}K)^T)I + \frac{\gamma}{n}K)$ where $\gamma = 2a + a^2$. Thus, each patch and atom are scaled and then normalized. This is the simplest implementation of the OMP with the new inner product. The original K-SVD, applied to $\sqrt{n} \times \sqrt{n} \times 3$ patches for colored image denoising.

Yu et al. (2013) and Xu et al. (2015) introduced the quaternion based sparse representation method (QSVD). It is observed that color image denoising methods propose by Elad and Aharon (2006a, b) and Mairal et al. (2008) does not capture the inter band correlation. Thus, they represent the RGB image as quaternion and its patch is represented as $y_c \in R^n$, where c is channel and n is patch dimension. The sparse model for color image be $y_c = [D_r;D_g;D_b]\alpha_c$; $c = r;g;b$ and final dictionary is the combination of three dictionaries. Obtaining a single dictionary from three dictionaries is a clustering task, which is non zero coefficients distribution in training samples. The color channels correlation is maintained using the quaternion based sparse model, which provides some constraints on dictionaries and coefficients, along with preserving the interrelationship among RGB channels. The patch of color image as quaternion is represented as $y = 0 + y_r i + y_g j + y_b k$. Here $y \in H^n$, is a n-dimension quaternion vector, and each component has 4 parts, and RGB image is represented as pure quaternion with the real part as zero. Similarly, dictionary and coefficient are denoted by $D = 0 + D_r i + D_g j + D_b k$ and $a_= a_0 + a_1 i + a_2 j + a_3 k$, respectively, resulting quaternion-based sparse representation model becomes $min_\alpha \mid\mid \alpha \mid\mid_0 ; s : t : y = Da$ here quaternion dictionary is represented by $D \in H^{n \times K}$, consisting of K pure quaternion atoms and $a \in H^K$ is a sparse quaternion vector. The objective function $\mid\mid a \mid\mid_0$ counts number of nonzero components in quaternion vector. This is sparse matrix with following constraints,

$$
\begin{aligned}
0 &= -D_r\alpha_1 - D_g\alpha_2 - D_b\alpha_3,\\
y_r &= D_r\alpha_0 + D_g\alpha_3 - D_b\alpha_2,\\
y_g &= -D_r\alpha_3 + D_g\alpha_0 + D_b\alpha_1,\\
y_b &= D_r\alpha_2 - D_g\alpha_1 + D_b\alpha_0.
\end{aligned}
$$

The equations above enforce explicit constraints on correlations among RGB channels where each y_c is linearly correlated with three channel dictionaries and coefficients lie in the null space of $[D_r;D_g;D_b]$. The quaternion based dictionary is trained where both dictionary and coefficients are considered as unknown variables. This process is formulated as

$$A.D = \underset{D}{argmin} \mid\mid Y - DA \mid\mid_2 - \lambda A,$$

where image patches set $Y^{n\times N} = \{y_i;1,2,\ldots,N\}$ and the quaternion-based dictionary, $D^{n\times K} = \{d_i;1 \le i \le K\}$ is composed of K atoms and sparse coefficients matrix $A^{K\times N} = \{\alpha_i;1 \le i \le N\}$. This dictionary is optimized using the *K-SVD* in quaternion algebra, called $K - QSVD$. It separates the training process into two steps, i.e, sparse coding stage and dictionary updating stage. During the sparse coding stage, a sparse coefficient matrix A is to be solved given the fixed D and *QSVD* problem is solved by orthogonal matching pursuit (OMP) algorithm for quaternion extension. The quaternion-based dictionary D is trained using the sparse coding solution. They further extend the model in Xu et al. (2015) and obtained a structured coefficient for preserving the correlation among color channels. By training the quaternion dictionary D in a proper way, the interrelationship of the three channels for color patches y can be well preserved. It imposes explicit linear correlation among the four channel dictionaries. A single quaternion-based dictionary is trained as an extension of the model in Lee et al. (2012) in which dictionary and coefficients are unknown variables. This process is called as training algorithm K-SVD in the quaternion form, which we call K-QSVD.

3.2.4 Results comparison

Table 3 shows the efficiency of the algorithms that are reported in terms of image quality measures (IQMs). These results are reported by the authors and they computed multiple IQMs for many color objects and noise levels. For simplicity, we only mentioned the one noise level which is also common in many of the methods reviewed here.

3.3 Artificial neural networks based methods

The artificial neural networks (ANN) is a self connected units for solving the complex real life problems iteratively. An ANN is a *massively parallel distributed processor that has a natural property for strong experimental knowledge and making it for practical use* (Haykin 1998). These networks are trained during the training phase and gained knowledge is represented by weights. The model training for feature extraction and optimal representation of data is the main step for deep learning algorithms. In ANN models, the data is transferred into useful knowledge by many interconnect layers. The ANN and deep learning methods are commonly used for image reconstruction, classification and object recognition tasks. Nowadays, these are also used for latent image restoration or image denoising tasks. Commonly developed deep learning algorithms for image restoration procedures are

Table 3 Sparse representation based color image denoising methods

Method	Images, noise ratio (%)	MPSNR	MSAM	MSSIM	MERGAS	MFSIM
MSt-SVD (Kong and Yang 2019)	Baboon, Lena, 1.5	39.3	0.19	0.94	62.4	–
HSID (Kong et al. 2019)	URBAN dataset, $\sigma = 0.3$	27.380	0.103	0.862	146.281	0.928
AFB-Voting (Shen et al. 2016)	House, Lena, 30	31.96, 31.46	–	–	–	–
TVF (Moreno et al. 2011)	Lena, Baboon	Proportional results are compared using PSNR	–	–	–	–
ITSReg (Xie et al. 2016)	Average for 32 images	37.78 ± 0.32	–	0.90 ± 0.07	77.35 ± 30.16	0.96 ± 0.02
MMBSR (Gai 2019)	Birds, Barbara, Pepper, 30	35.18, 35.26, 35.72	–	0.941 0.933 0.921	–	–
K-QSVD (Yu et al. 2013)	Barn, Athens, Blueeye, Bee, 20	27.34,28.26, 30.69, 28.24	–	0.67, 0.83, 0.84 , 0.81	–	–
KQSVD (Xu et al. 2015)	Barn, Athens, Blueeye, Bee, 20	29.41, 30.49, 33.07, 32.18	–	0.75, 0.89, 0.91, 0.88	–	–
Gaussian HSI (Hu et al. 2019)	Washington DC Mall, 0.5	39.900	–	.9134	–	–

based on different network architectures, however these ANN algorithms do not obtain the state of the art algorithms in image restoration and denoising, and some additional information can be used to improve the overall performance. In these methods, the following common terms are used:

- The Convolutional Neural Networks: The Convolutional Neural Networks (CNN) is a multilayered network model in which convolutional filters are implemented on nodes. These network models are effective for increasing capacity and flexibility for image characteristics, use of regularization and learning methods and including. Work on CNNs has been done since the late seventies (LeCun et al. 2015).
- Activation Function: The Neural network activation functions are an essential part of ANNs, and these functions defines the output of a layer in network. These functions also effect the computational cost, efficiency and convergence of the method and binary, linear , Sigmoid, max, Tanh are commonly known activation functions.
- Rectified Linear Unit: The Rectified Linear Unit (ReLU) is introduced by Krizhevsky et al. (2012a). This is a non-saturating nonlinearity. Mathematically $f(x) = \max(0;x)$.
- Sigmoid Function: The sigmoid function is defined as $\frac{1}{1+e^{-x}}$
- Batch Normalization: A simple CNN composed of activation function (Krizhevsky et al. 2012b), pooling operation (He et al. 2014), need some parameters to initiate and stopping optimization procedure (He et al. 2015), to optimize the error function and convolution kernels. The BN is used to speed up the computational process of large data set and prediction results (Ioffe and Szegedy 2015). It is the process through which the output of the previous activation layer is normalized, scaled or shifted by subtracting the mean and division by standard deviation.
- Residual learning: The residual learning is a method for learning high frequency components of a image in image restoration methods. It is used in many image processing fields such as dualdictionary learning by Zhang et al. (2012), image recognition ResNet (He et al. 2016), CNN network with residual learning for image denoising as in DnCNN (Zhang et al. 2017a).
- Receptive field: An output value of a pixel, in convolutional networks belongs to the particular part of an image. This image part is called receptive filed.

3.3.1 Contribution summaries

A deep learning method is introduced by Lehtinen et al. (2018). In this method, the loss function is formulated using negative likelihood, and a generalization of neural network regressor training. This loss function is minimized for every point of sample space. A multi-valued mapping with 12 loss functions is used for network training and out put is the plausible explanation. In the restoration procedure, L_1 loss function is implemented to recover median image.

In a nonlocal based CNN method introduced by Lefkimmiatis (2017), image regularization and potential functions are determined during the network training phase. The corrupted and ground truth data is used for training and image denoising is performed using a nonlocal regularization in patches. The patches are extracted and transformed using the $2D$ transformation. The KNN is used for patche selection and collaboration filtering is performed on these patches. The convolutional layers are used to compute a nonlocal operator $L_r x = (W\tilde{F}P_{i_r})X$, where $P_{i_r} = [P_{i_{r,1}}, P_{i_{r,2}}, P_{i_{r,3}}, \ldots, P_{i_{r,k}}]^T$ and $\tilde{F} \in R^{F.k \times p.k}$ is block matrix and its adjoint is $L_r^T x = (P_{i_r}^T \tilde{F}^T W^T$. The Gaussian redial functions are

used as potential function. In this method, each image channel is processed independently and performance is increased by using opponent color spaces instead of RGB spaces. The convolutional filters, weighted-sum layers and their transposes are shared for every image channel. This operation is computed once from noisy input and reused in all network stages, the algorithm is implemented for each channel parallel.

A deep learning method, hyperspectral imagery denoising by deep learning with trainable nonlinearity function (HDnTD) model is introduced (Xie and Li 2017). This methods is developed using the TNRD model. The TNRD model, the energy functional

$$\min_{u} E(u) = \sum_{i=1}^{N_k} \rho_i(k_i * u) + \frac{\lambda}{2} \mid\mid u - f \mid\mid_2^2, \tag{20}$$

where $*$ represents the convolution, is minimized. The Loss function is used to model weight matrix, bias matrix, and nonlinearity functions from the training data where linear filter and penalty functions are adjustable at each stage. The out put will become

$$u_t = u_{t-1} - \left(\sum_{i=1}^{N_k} \bar{k}_i^t * \phi_i^t(k_i^t * u_{i-1}) + \lambda^t(u_{i-1} - f) \right),$$

where N_k represents the linear filters and $\bar{k}$ is the rotation parameter. The one-stage TNRD model is equivalent to the two-layer feed-forward CNN and nonlinearity functions are trained for each stage. This is a new way of training the diffusion model. The nonlinearity function and information from training data are derived automatically in this model. In HDnTD model, the noisy HSI and ground truth data are prepared as the data and training process are labeled. Both training data and label of the HDnTD are normalized. The algorithm first extracts the overlapping cubic patches in the 38×38 pixels with 10 strides. The HDnTD is trained to extract the hierarchy of increasingly complex features where each stage performs convolution, nonlinearity and linear weighting operation without pooling. Initially, the noisy HSI is zero-padded and input into the trained model. For the next stage, the denoised result of previous stage is first padded and subsequently denoised in the same manner. To initialize procedure, the greedy learning algorithm is used at each stage and then quadratic loss function between clean and recovered images is adopted.

Zhang et al. (2019) introduced a method to denoise hyperspectral images (HSIs) with a spatial-spectral gradient network (SSGN). This noise is removed from both spatial and spectral domains by employing joint spatial and spectral gradient information. These gradients are defined as $G_x(m,n,k) = Y(m+1,n,k) - Y(m,n,k), G_y(m,n,k) = Y(m,n+1,k) - Y(m,n,k)$ and $G_z(m,n,k) = Y(m,n,k+1) - Ym,(n,k)$ where x, y, $z$are horizontal, vertical spatial , and spectral dimensions of current band. These gradients are used to estimate the residual noise of the relevant band and this process is repeated for each band and denoised image is reconstructed as $\hat{X}_k = F_l \circ \ldots \circ F_2 \circ F_1\{Y_k, G_x, G_y, G_z\}Y_k$ where F_l represents the lth multi-scale convolutional block operation of proposed model. This represents the feature map transformation from F_{l-1} to F_l. The $\hat{X}^k$ represents estimated denoised result of kth band. The multi-scale convolutional blocks are used to obtain diverse features for multi context information in spatial and spectral domains. The 3×3, 5×5 and 7×7 sizes kernels are used for spatial band, spatial gradient, and corresponding spectral gradient, this produces feature maps of 30 channels. The spatial term,

$$\xi_{spatial} = \frac{1}{2}\sum_{i=1}^{T} || Res_k^i - (Y_k^i - X_k^i) ||_2^2, \tag{21}$$

and spectral term,

$$\xi_{spectral} = \frac{1}{2}\sum_{i=1}^{T}\sum_{z=k-\frac{K}{2}}^{z=k+\frac{K}{2}} || \varphi_z^i - G_z^i ||_2^2 . \tag{22}$$

The method is trained by back-propagation algorithm using loss function, designed as a trainable parameters of spatial-spectral gradients. The loss function is defined as $\xi(\theta) = (1-\alpha)\xi_{spatial} - \alpha\xi_{spectral}$ where α is the penalty parameter for the trade-off between spatial and spectral items. The number of training patches $(T), Res_k$ is the kth band estimated residual spatial output and φ_z represents the estimated spectral gradient of bands $[k-\frac{K}{2} : k-1, k+1 : k+\frac{K}{2}]$.

Yuan et al. (2019) utilized spatial-spectral gradients information for denoising HSIs, in which Deep CNN (HSID-CNN) is employed for data training. The method deals with flexible band numbers and method takes one single spatial band (2D) for denoising, and its adjacent spectral bands as auxiliary data. A nonlinear end-to-end mapping between noisy data and original data is defined by CNN, where $y_{spatial}$ [Eq. (21)] and $y_{spectral}$ [Eq. (22)] are simulated. The multi-scale feature are captured by employing multiple convolutional operators with kernel sizes and six outputs of feature maps are concatenated into a single 120-channel feature map. The deep networks are implemented to extract essential features and non-linearity among them. The spatial distribution of residual feature maps are sparse, due to this reason, the gradient descent process is transferred to a smoother hyper-surface of loss and residual image. The stability and efficiency of training procedure is ensured by using residual learning. The residual noise $\varphi = \hat{x} - y_{spatial}$ where $\hat{x}$ is the original clean band is used in image reconstruction. In HSID-CNN, the training images are taken as collection $T = \{x^i, y^i_{spatial}, y^i_{spectral}\}$, where y are noisy HSI as the spatial, spectral data and i is corresponding noisy adjacent cube as the spectral data, and x^i is the clean HSI as the label. Setting Θ as network trainable parameters, the mean-squared error (MSE) is utilized as loss function,

$$\Theta = \frac{1}{2T}\sum_{l=1}^{T} || NET(y^i_{spatial}, y^i_{spectral}, \Theta) - \varphi^i || .$$

The concatenation of convolution operators is used to combine multiple features, called multiple skip-connections in Wei et al. (2017). This is mathematically described as $f_c = Concat\{f_3, f_5, f_7, f_9\}$ where $f_3, f_5 f_7 f_9$ represents the different-level feature, respectively. The concatenated layer f_c is employed to fuse these feature. Then $\varphi = W_c * f_c + b_c$ where W_c and b_c stand for the weight parameters and bias parameter of the last convolutional layer.

The hierarchical residual learning method proposed by Shi et al. (2019) is the extension of method proposed by Kim et al. (2016), in which the network depth is decreased with the improvement in output. The denoising algorithms do not depend on depth of network, as a counter example is provided in Bevilacqua et al. (2012). In this method, the residual learning by identity mapping is replaced with residual map using more than one receptive fields defined as $f_i(y) = max(0, W_i * Y + B_i)$ and $F(Y) =$ Fusion $(f_1(y), f_2(y), f_3(y), \ldots, f_n(y))$. This map is optimized progressively in hierarchical residual learning. All these information are fused using Concat layer. The image noise is estimated by hierarchical residual learning. This

network consist of three subnetworks, called feature extraction, inference and fusion networks. The feature extraction network learns the effective feature maps from the noisy image, noise is estimated by inference network and finally, all these estimated noise maps are fused to get the final estimated noise by a fusion net. The feature extraction sub-network is a single layered network which takes in noisy image and defines the feature map by convolution. The inference network can extract information for a nonflexible single level information and for multi scale information a cascade structure is used and this structure is supportive for multiple residuals learning idea. The fusion sub-network fuses all noise maps and finally image noise estimation is obtained. The concat layer group these noise maps together into a high dimensional noise feature map. The mean square error is adopt as the loss function and objective is to minimize this function. For training data set $\{x_i, y_i\}_{i=1}^{n}$ where x_i and y_i are noise image and its corresponding noise map respectively. The optimization of jth inference sub-network defined as $F_1^j(\theta) = \frac{1}{2n}\sum_{i=1}^{n} \| F_1^j(x_i, \theta_j) - y_i \|$ where θ and $F_1^j(x_i, \theta_j)$ are parameters and noise map of the jth inference sub-network. For fusion sub-network, $L_R^j(\theta) = \frac{1}{2n}\sum_{i=1}^{n} \| F_R^j(x_i, \theta_j) - y_i \|_F^2$ where θ_j and $F_R^j(x_i, \theta)$ are the network parameters for training and final estimated noise map respectively. Finally, the loss function becomes $L(\theta) = \alpha_R L_R(\theta) + \sum_{i=1}^{N} \alpha_i L_i(\theta_i)$, where α_R and α_i denote the importance of corresponding loss functions. This method extracts multiple scale information learning and robust to fault tolerance.

The batch-renormalization denoising network (BRDNet) method for color image denoising is introduced by Tian et al. (2020b). In this method, the network is based on two subnetworks, called upper and lower networks. The lower network includes BRN, RL, and dilated convolutions. The contextual information are captured at the 2–8 and 10–15 layers of lower network using dilated convolutions. Whereas, data is normalized at first, ninth and sixteenth layers through BRN. The upper network consist of is 17 layers, which are two types. Layers 1–16 are Conv + BRN + ReLU, Convolution, batch renormalization, and rectified linear units, respectively. Where as 17th layer is Conv., except for the first and last layers, each layer of size $64 \times 3 \times 3 \times 64$. The sizes of first and last layers are $c \times 3 \times 3 \times 64$ and $64 \times 3 \times 3 \times c$, respectively, where c denotes channels, for RGB images $c = 3$. The loss function is modeled using MSE.i.e., $f(y) = y - x$ where x is clean image, y be a noisy image and $f(y)$ is residual image, obtained from the RL in the BRDNet method. The optimal parameters can be obtained by minimizing the following loss function with Kingma and Ba (2014), $l(\theta) = \frac{1}{2N}\sum_{i=1}^{N} \| f(y_i, \theta) - (y_i - x_i) \|^2$, where noisy-image patches are denoted by N and θ is the model parameter. The objective function is relevant to the noisy image y, clean image x, and residual image $f(y)$. The use of image patches can handle inhomogeneous image regions and save memory and computational cost. This method increases width by introducing two different sub-networks and reduce depth of the network and obtain more features. Two networks and dilated convolution is combined to increase the efficiency of algorithm.

Li et al. (2020) proposed a DRCNN model, in which loss function is defined using the mean square error,

$$l(\theta) = \frac{1}{2N}\sum_{i=1}^{N} \| [y_i - n(y_i;\theta) + D'(y, \theta)] - x_i) \|^2,$$

where D', y_i and θ are the information lost during the denoising process, noisy image and learning parameter respectively. The $n(y_i;\theta)$ is the residual image and v_i stands for AWGN and x_i is a clean image. The input of DRCNN is noisy image y and the output is,

$$Y = y - n(y;\theta) + D'(y;\theta). \tag{23}$$

The learning process in Eq. (23) includes two mappings. First mapping concerns with the generation of residual image $n(y;\theta)$ and second mapping is relearning image detail information D'. This network consist of 30 convolution layers using 3×3 filter size and stride is one in each layer and two modules. The first, generation module (GM) used for generating IFM, includes convolution layers from 1 to $d_0 + 1$. There are c and 64 output channels of the $d_0 + 1$th layer and other layers respectively. The image detail retaining module (DRM) is used for retaining image detail and this includes convolutional layers from $d_0 + 2$ to 30. There are c output channels of the 30th layer. The value of c is the same as the number of channels of the input noisy image. In this work, d_0 is set to 19. The DRCNN does not include BN layer and method learns noise and image details simultaneously.

Guo et al. (2018), developed a high performance deep CNN algorithm, called convolutional blind denoising network (CBDNet). It composed of two subnetworks, estimation subnetwork CNN_E and denoising subnetwork CNN_D. The network CNN_E consist of five convolutional layers without pooling and batch normalization. The synthetic images and noise is used for network training. The CNN_E considers noisy image y as input and produces estimated noise level map $\hat{\sigma}(y) = F_E(y;W_E)$, where W_E is the parameter of CNN_E. Both, input and out put of CNN_E are inputs to second subnetwork CNN_D. Its output is $\hat{X} = F_D(y;\hat{\sigma}(y);W_D)$, where W_D is the network parameter. The noise level $\hat{\sigma}(y)$ is adjustable before the use in subnetwork CNN_D. The ReLU nonlinear function is used as a weight and U net architecture is implemented in CNN_D with 16 layered network. The strider and transpose convolutions are introduced for exploiting multi-scale information as well as enlarging receptive field. The ReLU nonlinearity is applied on every layer except the output layer and batch normalization is used for Gaussian denoising.

A multi-level wavelet-CNN (MWCNN) based algorithm is presented by Liu et al. (2018). In 2D discrete wavelets, four convolutional filters are used resulting the four sub sampled images as output. The original image is reconstructed by inverse wavelet transform (IWT). This process is continued upto three or more levels and a CNN block is added in between two layers of Wavelet sub-sampling. At decomposition stage, four pre-defined filters are deployed to each (subband) image, and downsampling is then adopted as pooling operator. The wavelet transformation also preserve the texture information of the input image. At reconstruction, a four layered FCN without pooling is considered as CNN block that takes all subband images as input and every CNN block consist of 3×3 convolution filters, BN, and ReLU operations except the last layer. The down sampling, upsampling and max pooling and up convolution steps are modified in U net and used in MWCNN and summation is used for combination of features. This network composed of 24 layers.

Zhang et al. (2017b) proposed a fast speed and flexible neural network (FFDNET) model. A tunable noise level MAP is used as input with a reversible down sampling operator. The input image $W \cdot H \cdot C$ (for colored image, $C = 3$) is reshaped with $\frac{W}{2} \cdot \frac{H}{2} \cdot 4C$ and orthogonal initialization method for convolution filters is implemented. The resulting tensors $\frac{W}{2} \times \frac{H}{2} \times 4C + 1$ used as an input to CNN. The CNN consist of a 3×3 Convolution, ReLU, and BN layers. A zero padding is introduced at the last layer. The estimated denoised image of size $W \times H \times C$ is reconstructed by up sampling operator. This network consist of 12 convolution layers and output is a 96 dimensional feature vector. The various noise levels in images are handled using the model based methods. The image dimensions and dimensions of noise level are equalized by image patching, for colored images $3D$ Gaussian noise is used. Alternatively, A 15 layered network is used with 3×3 convolutional networks with a receptive field of 62×62. The denoising results are improved by increasing the size receptive field. The orthogonal regularization methods is used better noise minimization and preserving image details.

A CNN based model is proposed by Zhang et al. (2017a). For this purpose, the VGG network model is modified. The network layers are adjustable according to the size of patches. The residual learning and batch normalization are used for model training. A convolution filters of size 3×3 without pooling functions are implemented. The patch size is directly proportional to noise ratio. The effective patch size is analyzed by fixing the noise level $\delta = 25$. The receptive filed of DnCNN is fixed to 35 with the depth of 17 layers to deal the Gaussian noise and a larger receptive field with 20 layered depth is used commonly in image denoising tasks. The objective of the algorithm is to determined the mapping function $F(y) = x$ to predict the latent clean image using the noisy observation $y = x + v$. The residual mapping $R(y) \approx v$,and $x = y - R(y)$ are determined by residual learning formulation and Mean square error is used as a loss function. This network has three layers, first layer consist of Conv+ReLU function with 64 filters of size $3 \times 3 \times c$ and 64 dimensional feature map is generated. At the second layer, a 64 convolution filters of size $3 \times 3 \times 64$ is used. To speed up and image denoising process, batch normalization is performed and this is followed by ReLU operator. At the last layer, the image is reconstructed using the c filters of size $3 \times 3 \times 64$. The image boundary is treated using zero padding and optimal points are achieved using the Gradient descent algorithm with momentum and Adam optimization.

The methods for Pixel-Distribution Prior for image denoising using CNN was introduced by Liu and Fang (2017). These methods have 5 layers of 128 convolution filters. The statistical features, mean and variance are employed for training data at training phase is determined by BN. The WIN5-RB has two types of layers with two different colors, first four layers are consist of Conv + BN + ReLU, BN is added between Conv and ReLU. where as, the ReLU is skipped from last layer. The application of BN at every layer is helpful for preserving prior image information during training phase. The image is reconstructed using a 7×7 filter. Image input and out put layers are also connected in the image recovery phase.

3.3.2 Results comparison

The image denoising algorithms developed using the artificial neural network techniques are trained using a test data set. For this purpose, many bench marking test data sets are developed and these algorithms are implemented on these data sets. These data sets include:

- Noise Clinic: The data set Noise Clinic (NC12) is introduced by Lebrun et al. (2015). This data set contains 12 noisy images along with the ground truth images.
- Darmstadt Noise Dataset: Plötz and Roth (2017) introduced a benchmarking data set called Darmstadt Noise Dataset (DND). It contains 50 pairs of real noisy and corresponding noise free images.
- Kodak: Kodak or Kodak24 (Franzen 1999) is a true color 24 bits pixel images data set introduced by the Eastman Kodak Company for unrestricted usage. It consist of 24 images.
- The Berkeley Segmentation Dataset (BSD) is introduced by Martin et al. (2001). It is a collection of 12,000 hand-labeled segmentations of 1000 Corel dataset images from 30 human subjects. In algorithms, number of images are chosen for verification of algorithms. These numbers are indicated with BSD such as BSD68, BSD200, BSD300, etc. Its color version is also known as CBSD.

Table 4 Comparison ANN and deep learning methods for color image denoising

Method	Noise type and level	Image data	MMSE	PSNR	MSSIM
NC (Lebrun et al. 2015)	WG, 30, 40	Real images	–	(29.22, 29.94)	–
NLCCNN (Lefkimmiatis 2017)	AG, 25, 50	Berkeley images	–	(30.96, 27.64)	–
FFDNET (Zhang et al. 2017b)	AWG, 50, 75	Real images, CBSD68, Kodak24, Mc master	(27.96, 26.24), (28.98, 27.27), (29.18, 27.33)	–	–
DCNN (Zhang et al. 2017a)	Gaussian, 25, 50	BSD 68, Set5, set14, BSDB100, Urban 100	(29.02, 26.10), (33.75, 31.41), (29.81, 28.04), (28.85, 27.29), (30.74, 25.20)	–	(0.819, 0.708), (0.922, 0.884), (0.832, 0.767), (0.798, 0.725), (0.828, 0.752)
LPWCDen (Liu and Fang 2017)	Gaussian, 50	BSD200	WIns= 23.88, WINS-R =24.70, WINS-RB= 26.92	–	Wins= 0.5858, WINS-R =0.6640, WINS-RB= 0.8254
N2N (Lehtinen et al. 2018)	Gaussian(25), Poisson (30) Bernoulli (0.5)	Kodak, BSD300, Set14	(31.63, 30.59, 31.85)	–	–
MWCNN (Liu et al. 2018)	Gaussian 25,50	Set12, BSD68, Urban100	(30.79,27.74), (29.41, 26.53), (30.60, 27.42)	–	(0.871, 0.866), (0.836, 0.737), (0.903, 0.837)
CBDen (Guo et al. 2018)	AWGN	NC12, DND, Nam	38.06	–	0.9421
DnCNN (Cha and Moon 2019)	30,50	Set13, BSD68	28.64, 28.41	–	–
ANNNLE (Turajlic et al. 2019)	AWGN, 40, 45, 50	Barbara, Car, Lena, Town	3.10 to 3.26	–	–
SSDA (Xie et al. 2012)	WGN, Salt & Pepper, 50	Lena, Woman, Yachet	–	24.19 ± 1.39	–
HDnTD (Xie and Li 2017)	Gaussian, 25	Cave,Harvard data sets,	FSIM= 0.9622, ERGAS= 43.1629, SAM= 0.0629	43.022	0.9595
SSGN (Zhang et al. 2019)	Gaussian, Strip,	Hydice, Hyperion EO,Indian Pines	–	34.37	0.982
HSID-CNN (Yuan et al. 2019)	Gaussian, random, 25	Washington DC Mall, AVIRIS Indian Pines, ROSIS	MSA (3.3156 ± 0.0017)	36.804 ± 0.0029	0.9895 ± 0.0001
HRLNet (Shi et al. 2019)	GN , 25	Set5, set12,BSD68	–	(31.80, 30.46, 29.14)	(0.8766, 0.8368, 0.8238)

Table 4 (continued)

Method	Noise type and level	Image data	MMSE	PSNR	MSSIM
BRDNet (Tian et al. 2020b)	AWGN, 50	CBSD68, Kodak 24, McMaster	–	28.16, 29.22,29.52	–
DRCNN (Li et al. 2020)	Gaussian,50	Set12, BSD68	–	27.290, 26.29	0.8190, 0.7323
CTCNN (Lyu et al. 2020)	Gaussian, 50	Set12, BSD68, 50	–	27.69, 26.56	–
DDRN (Wang et al. 2017a)		BSD68, 50	–	27.93	–
ADNet (Tian et al. 2020a)	50	BSD68, Kodak 24, McMaster	–	26.29, 29.10, 29.36	–

- Urban100 : This data set is introduced by Huang et al. (2015), containing 100 HR images.
- Set: A simple and self arranged dataset for experiments such as set of 5 images, namely Set5 (Bevilacqua et al. 2012) and Set of 12 images is called Set12 (Zhang et al. 2017a) and set of 14 images is known as Set14 (Zeyde et al. 2012). These are datasets arranged individually for experiments.
- McMaster: The dataset is developed by McMaster University (Wu 2011), Canada, in collaboration with some industry partners for developing new CDM methods. This dataset has 8 high resolution images.

Table 4 presents a comparison of ANN and deep learning based color image denoising models. Note that these methods are applied with multiple choice of noise levels and noise types. The results are compared using the statistical measures PSNR (db), MSE and SSIM. The results are compared for multiple noise levels, however, here these are mentioned for selected noise levels, the parenthesis are used to express the MSE, PSNR or SSIM results for multiple noise levels and for multiple data sets these are separated by commas. The measures which are not described by authors are not mentioned and left blank as dashed lines. In some methods, results are represented for average of the total experiments. In these cases, the measures are represented as a range. As can be seen, the deep learning models outperform the traditional PDE and low rank, sparse representation based methods considered previously. However, these deep learning models typically use large scale training datasets and the computational complexity compared to PDEs or sparse representation models are much higher.

4 Conclusion

In this review, we reviewed image denoising and restoration with partial differential equations (PDEs), low rank and sparse representation models, and recent deep learning methods based on artificial neural networks (ANNs). We observed that the deep learning and ANN methods of image denoising deals with the spacial class of color images, including multispectral and hyperspectral images that contain more than three channels. In recent years, much works are devoted for ANN based methods whereas less attention have been paid to traditional mathematical models, i.e., PDEs and sparse representation models for color image denoising methods.

Funding Funding was provided by National Center for Advancing Translational Sciences (Grant No. U2CTR002818), National Heart, Lung, and Blood Institute (Grant No. U24HL148865), National Institute of Allergy and Infectious Diseases (Grant No. U01AI150748), Cincinnati Children's Hospital Medical Center (Grant No. Advanced Research Council (ARC) Grant 2018-2020).

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

Ambrosio L, Tortorelli VM (1990) Approximation of functional depending on jumps by elliptic functional via gamma-convergence. Commun Pure Appl Math 43(8):999–1036. https://doi.org/10.1002/cpa.3160430805

Astrom F, Schnorr C (2017) A geometric approach for color image regularization. Comput Vis Image Underst 165:43–59. https://doi.org/10.1016/j.cviu.2017.10.013

Bar L, Brook A, Sochen N, Kiryati N (2005) Color image deblurring with impulsive noise. In: Paragios N, Faugeras O, Chan T, Schnorr C (eds) Variational, geometric, and level set methods in computer vision. Springer, Berlin, pp 49–60

Barash D (2001) One-step deblurring and denoising color images using partial differential equations. Techcical report HPL2000, HP Labortries

Bevilacqua M, Roumy A, Guillemot C, Morel MA (2012) Low-complexity single-image super-resolution based on nonnegative neighbor embedding. In: Proceedings of the British machine vision conference. BMVA Press, pp 135.1–135.10. https://doi.org/10.5244/C.26.135

Blomgren P, Chan TF (1998) Color TV: total variation methods for restoration of vector-valued images. IEEE Trans Image Process 7(3):304–309. https://doi.org/10.1109/83.661180

Boncelet C (2009) Chapter 7—image noise models. In: Bovik A (ed) The essential guide to image processing. Academic Press, Boston, pp 143–167. https://doi.org/10.1016/B978-0-12-374457-9.00007-X

Buades A, Coll B, Morel J (2005) A review of image denoising algorithms, with a new one. Multiscale Model Simul 4(2):490–530. https://doi.org/10.1137/040616024

Cha S, Moon T (2019) Fully convolutional pixel adaptive image denoiser. In: Proceedings of the IEEE international conference on computer vision, pp 4160–4169

Chan T, Shen J (2001) Variational restoration of nonflat image features: models and algorithms. SIAM J Appl Math 61(4):1338–1361. https://doi.org/10.1137/S003613999935799X

Chan TF, Kang SH, Shen J (2001a) Total variation denoising and enhancement of color images based on the CB and HSV color models. J Vis Commun Image Represent 12(4):422–435. https://doi.org/10.1006/jvci.2001.0491

Chan TF, Osher S, Shen J (2001b) The digital TV filter and nonlinear denoising. IEEE Trans Image Process 10:231–41. https://doi.org/10.1109/83.902288

Elad M, Aharon M (2006a) Image denoising via learned dictionaries and sparse representation. In: 2006 IEEE computer society conference on computer vision and pattern recognition (CVPR), vol 1, pp 895–900

Elad M, Aharon M (2006b) Image denoising via sparse and redundant representations over learned dictionaries. IEEE Trans Image Process 15(12):3736–3745

Franzen R (1999) Kodak lossless true color image suite. Available online at http://r0k.us/graphics/kodak/. Accessed 03 Mar 2021

Fuchs M, Muller J, Tietz C, Weickert J (2018) Convex regularization of multi-channel images based on variants of the TV-model. Complex Var Elliptic Equ 63(7–8):976–995. https://doi.org/10.1080/17476933.2017.1386181

Gai S (2019) Color image denoising via monogenic matrix-based sparse representation. Vis Comput 2019(35):109–122

Goyal B, Dogra A, Agrawal S, Sohi BS, Sharma A (2020) Image denoising review: from classical to state-of-the-art approaches. Inf Fusion 55:220–244. https://doi.org/10.1016/j.inffus.2019.09.003

Grecova S, Morillas S (2016) Perceptual similarity between color images using fuzzy metrics. J Vis Commun Image Represent 34:230–235. https://doi.org/10.1016/j.jvcir.2015.04.003

Gu S, Zhang L, Zuo W, Feng X (2014) Weighted nuclear norm minimization with application to image denoising. In: 2014 IEEE conference on computer vision and pattern recognition, pp 2862–2869

Guichard F, Moisan L, Morel JM (2002) A review of P.D.E. models in image processing and image analysis. J Phys IV 12:137–154. https://doi.org/10.1051/jp42002006

Guo S, Yan Z, Zhang K, Zuo W, Zhang L (2018) Toward convolutional blind denoising of real photographs. CoRR arxiv: abs/1807.04686

Hassan M, Bhagvati C (2012) Structural similarity measure for color images. Int J Comput Appl 43:7–12. https://doi.org/10.5120/6169-8590

Haykin S (1998) Neural networks: a comprehensive foundation, 2nd edn. Prentice Hall PTR, Upper Saddle River

He J, Zhang X, Ren S, Sun J (2014) Spatial pyramid pooling in deep convolutional networks for visual recognition. In: Fleet D, Pajdla T, Schiele B, Tuytelaars T (eds) ECCV. Springer, Cham, pp 346–361

He K, Zhang X, Ren S, Sun J (2015) Delving deep into rectifiers: surpassing human-level performance on imagenet classification. In: IEEE international conference on computer vision (ICCV), vol 1502. https://doi.org/10.1109/ICCV.2015.123

He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: 2016 IEEE conference on computer vision and pattern recognition (CVPR), pp 770–778

Hu Z, Huang Z, Huang X, Luo F, Ye R (2019) An adaptive nonlocal Gaussian prior for hyperspectral image denoising. IEEE Geosci Remote Sens Lett 16(9):1487–1491

Huang J, Singh A, Ahuja N (2015) Single image super-resolution from transformed self-exemplars. In: 2015 IEEE Conference on computer vision and pattern recognition (CVPR), pp 5197–5206. https://doi.org/10.1109/CVPR.2015.7299156

Ioffe S, Szegedy C (2015) Batch normalization: accelerating deep network training by reducing internal covariate shift. In: 32nd International Conference on Machine Learning, PMLR 37. pp 448–456

Kim S (2006) PDE-based image restoration: a hybrid model and color image denoising. IEEE Trans Image Process 15(5):1163–1170. https://doi.org/10.1109/TIP.2005.864184

Kim J, Lee JK, Lee KM (2016) Accurate image super-resolution using very deep convolutional networks. In: IEEE conference on computer vision and pattern recognition (CVPR), pp 1646–1654

Kingma D, Ba J (2014) Adam: a method for stochastic optimization. In: International conference on learning representations

Kolda TG, Bader BW (2009) Tensor decompositions and applications. SIAM Rev 51(3):455–500. https://doi.org/10.1137/07070111X

Kong Z, Yang X (2019) Color image and multispectral image denoising using block diagonal representation. IEEE Trans Image Process 28(9):4247–4259

Kong X, Zhao Y, Xue J, Chan JCW (2019) Hyperspectral image denoising using global weighted tensor norm minimum and nonlocal low-rank approximation. Remote Sens 11:2281

Krizhevsky A, Sutskever I, Hinton GE (2012a) Imagenet classification with deep convolutional neural networks. In: Proceedings of the 25th international conference on neural information processing systems, vol 1, NIPS. USA, pp 1097–1105

Krizhevsky A, Sutskever I, Hinton GE (2012b) Imagenet classification with deep convolutional neural networks. In: Pereira F, Burges CJC, Bottou L, Weinberger KQ (eds) Advances in neural information processing systems, vol 25. Curran Associates, Inc., pp 1097–1105. http://papers.nips.cc/paper/4824-imagenet-classification-with-deep-convolutional-neural-networks.pdf

Lebrun M, Colom M, Morel JM (2015) The noise clinic: a blind image denoising algorithm. Image Process On Line 5:1–54. https://doi.org/10.5201/ipol.2015.125

LeCun Y, Bengio Y, Hinton G (2015) Deep learning. Nature 521(7553):436–444. https://doi.org/10.1038/nature14539

Lee SH, Choi JY, Ro YM, Plataniotis KN (2012) Local color vector binary patterns from multichannel face images for face recognition. IEEE Trans Image Process 21(4):2347–2353

Lefkimmiatis S (2017) Non-local color image denoising with convolutional neural networks. In: IEEE conference on computer vision and pattern recognition (CVPR), pp 5882–5891. https://doi.org/10.1109/CVPR.2017.623

Lehtinen J, Munkberg J, Hasselgren J, Laine S, Karras T, Aittala M, Aila T (2018) Noise2noise: Learning image restoration without clean data. CoRR arxiv: abs/1803.04189

Li X, Xiao J, Zhou Y, Ye Y, Lv N, Wang X, Wang S, Gao S (2020) Detail retaining convolutional neural network for image denoising. J Vis Commun Image Represent. https://doi.org/10.1016/j.jvcir.2020.102774

Liu P, Fang R (2017) Learning pixel-distribution prior with wider convolution for image denoising. CoRR arxiv: abs/1707.09135

Liu P, Zhang H, Zhang K, Lin L, Zuo W (2018) Multi-level wavelet-CNN for image restoration. CoRR arxiv: abs/1805.07071

Luo M, Cui G, Rigg B (2001) The development of the CIE 2000 colour-difference formula: Ciede 2000. Color Res Appl 26:340–350. https://doi.org/10.1002/col.1049

Lyu Z, Zhang C, Han M (2020) A nonsubsampled countourlet transform based CNN for real image denoising. Signal Process Image Commun 82(115):727. https://doi.org/10.1016/j.image.2019.115727

Ma Z, Wu HR, Feng D (2007) Fuzzy vector partition filtering technique for color image restoration. Comput Vis Image Understand 107(1):26–37. https://doi.org/10.1016/j.cviu.2006.11.017

Mairal J, Elad M, Sapiro G (2008) Sparse representation for color image restoration. IEEE Trans Image Process 17(1):53–69

Martin D, Fowlkes C, Tal D, Malik J (2001) A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics. In: Proceedings

eighth IEEE international conference on computer vision. ICCV 2001, vol 2, pp 416–423. https://doi.org/10.1109/ICCV.2001.937655

Medioni G, Lee MS, Tang CK (2000) A computational framework for segmentation and grouping. Elsevier Science B.V., Amsterdam. https://doi.org/10.1016/B978-044450353-4/50002-0

Melange T, Nachtegael M, Kerre EE (2011) Fuzzy random impulse noise removal from color image sequences. IEEE Trans Image Process 20(4):959–970. https://doi.org/10.1109/TIP.2010.2077305

Moreno JC, Prasath VBS, Neves JC (2016a) Color image processing by vectorial total variation with gradient channels coupling. Inverse Probl Imaging 10(2):461–497. https://doi.org/10.3934/ipi.2016008

Moreno JC, Prasath VS, Santos G, Proença H (2016b) Robust periocular recognition by fusing sparse representations of color and geometry information. J Signal Process Syst 82(3):403–417. https://doi.org/10.1007/s11265-015-1023-3

Moreno R, Garcia MA, Puig D, Julia C (2011) Edge-preserving color image denoising through tensor voting. Comput Vis Image Understand 115(11):1536–1551. https://doi.org/10.1016/j.cviu.2011.07.005

Mukhopadhyay S, Mandal JK (2014) A fuzzy switching median filter of impulses in digital imagery (FSMF). Circuits Syst. Signal Process 33(7):2193–2216. https://doi.org/10.1007/s00034-014-9739-z

Mumford DB, Shah J (1989) Optimal approximations by piecewise smooth functions and associated variational problems. Communications on Pure and Applied Mathematics 42(5):577–685. https://doi.org/10.1002/cpa.3160420503

Pal C, Chakrabarti A, Ghosh R (2015) A brief survey of recent edge-preserving smoothing algorithms on digital images. CoRR arxiv: abs/1503.07297

Perona P, Malik J (1990) Scale-space and edge detection using anisotropic diffusion. IEEE Trans Pattern Anal Mach Intell 12(7):629–639. https://doi.org/10.1109/34.56205

Plötz T, Roth S (2017) Benchmarking denoising algorithms with real photographs. In: 2017 IEEE conference on computer vision and pattern recognition (CVPR), pp 2750–2759. https://doi.org/10.1109/CVPR.2017.294

Prasath VBS, Singh A (2010) Multispectral image denoising by well-posed anisotropic diffusion scheme with channel coupling. Int J Remote Sens 31(8):2091–2099. https://doi.org/10.1080/01431160903260965

Prasath VS (2011) Weighted Laplacian differences based multispectral anisotropic diffusion. In: 2011 IEEE international geoscience and remote sensing symposium, pp 4042–4045. IEEE. https://doi.org/10.1109/IGARSS.2011.6050119

Prasath VS, Moreno JC (2018) On convergent finite difference schemes for variational–PDE-based image processing. Comput Appl Math 37(2):1562–1580. https://doi.org/10.1007/s40314-016-0414-9

Prasath VS, Vorotnikov D (2014) Weighted and well-balanced anisotropic diffusion scheme for image denoising and restoration. Nonlinear Anal Real World Appl 17:33–46. https://doi.org/10.1016/j.nonrwa.2013.10.004

Rafsanjani HK, Sedaaghi MH, Saryazdi S (2017) An adaptive diffusion coefficient selection for image denoising. Digit Signal Process 64:71–82. https://doi.org/10.1016/j.dsp.2017.02.004

Shah J (1996) A common framework for curve evolution, segmentation and anisotropic diffusion. In: Proceedings CVPR IEEE computer society conference on computer vision and pattern recognition, pp 136–142. https://doi.org/10.1109/CVPR.1996.517065

Shen Y, Han B, Braverman E (2016) Adaptive frame-based color image denoising. Appl Comput Harmonic Anal 41(1):54–74

Shi W, Jiang F, Zhang S, Wang R, Zhao D, Zhou H (2019) Hierarchical residual learning for image denoising. Signal Process Image Commun 76:243–251. https://doi.org/10.1016/j.image.2019.05.007

Sochen N, Kimmel R, Malladi R (1998) A general framework for low level vision. IEEE Trans Image Process 7(3):310–318. https://doi.org/10.1109/83.661181

Tian C, Xu Y, Li Z, Zuo W, Fei L, Liu H (2020a) Attention-guided CNN for image denoising. Neural Netw 124:117–129. https://doi.org/10.1016/j.neunet.2019.12.024

Tian C, Xu Y, Zuo W (2020b) Image denoising using deep CNN with batch renormalization. Neural Netw 121:461–473. https://doi.org/10.1016/j.neunet.2019.08.022

Tschumperle D (2006) Fast anisotropic smoothing of multi-valued images using curvature-preserving PDE's. Int J Comput Vis 68:65–82. https://doi.org/10.1007/s11263-006-5631-z

Tschumperle D, Deriche R (2007) Anisotropic diffusion partial differential equations for multichannel image regularization: framework and applications. In: Hawkes PW (ed) Advances in imaging and electron Physics, vol 145. Elsevier, Amsterdam, pp 149–209

Turajlic E, Begovic A, Skaljo N (2019) Application of artificial neural network for image noise level estimation in the SVD domain. Electronics 8(2):20

Venetsanopoulos AN, Plataniotis KN (2000) Adaptive filters for color image processing: a survey. In: 2000 10th European signal processing conference, pp 1–4

Wald L (2002) Data fusion. Definitions and Architectures—Fusion of Images of Different Spatial Resolutions. Presses de l'Ecole, Ecole des Mines de Paris, Paris. https://hal-mines-paristech.archives-ouvertes.fr/hal-00464703. ISBN 2-911762-38-X

Wang Z, Simoncelli EP, Bovik AC (2003) Multiscale structural similarity for image quality assessment. In: The thrity-seventh Asilomar conference on signals, systems computers, vol 2, pp 1398–1402. https://doi.org/10.1109/ACSSC.2003.1292216

Wang T, Sun M, Hu K (2017a) Dilated residual network for image denoising. CoRR arxiv: abs/1708.05473

Wang Y, Yang Y, Chen T (2017b) Spectral-spatial adaptive and well-balanced flow-based anisotropic diffusion for multispectral image denoising. J Vis Commun Image Represent 43:185–197. https://doi.org/10.1016/j.jvcir.2017.01.005

Wei Y, Yuan Q, Shen H, Zhang L (2017) Boosting the accuracy of multispectral image pansharpening by learning a deep residual network. IEEE Geosci Remote Sens Lett 14(10):1795–1799

Wu X (2011) Color demosaicking by local directional interpolation and nonlocal adaptive thresholding. J Electron Imaging 20(2):023,016. https://doi.org/10.1117/1.3600632

Wyszecki G, Stiles W (2000) Color science: concepts and methods, quantitative data and formulae. In: Color science: concepts and methods, quantitative data and formulae, 2nd edn. Wiley, New York

Xie J, Xu L, Chen E (2012) Image denoising and inpainting with deep neural networks. In: Pereira F, Burges CJC, Bottou L, Weinberger KQ (eds) Advances in neural information processing systems, vol 25. Curran Associates, Inc., pp 341–349. http://papers.nips.cc/paper/4686-image-denoising-and-inpainting-with-deep-neural-networks.pdf

Xie Q, Zhao Q, Meng D, Xu Z, Gu S, Zuo W, Zhang L (2016) Multispectral images denoising by intrinsic tensor sparsity regularization. In: 2016 IEEE conference on computer vision and pattern recognition (CVPR), pp 1692–1700. https://doi.org/10.1109/CVPR.2016.187

Xie W, Li Y (2017) Hyperspectral imagery denoising by deep learning with trainable nonlinearity function. IEEE Geosci Remote Sens Lett 14(11):1963–1967

Xu Y, Yu L, Xu H, Zhang H, Nguyen T (2015) Vector sparse representation of color image using quaternion matrix analysis. IEEE Trans Image Process 24(4):1315–1329

Yu L, Xu Y, Xu H, Zhang H (2013) Quaternion-based sparse representation of color image. In: IEEE International conference on multimedia and expo (ICME), pp 1–7

Yuan Q, Zhang Q, Li J, Shen H, Zhang L (2019) Hyperspectral image denoising employing a spatial-spectral deep residual convolutional neural network. IEEE Trans Geosci Remote Sens 57(2):1205–1218

Zeyde R, Elad M, Protter M (2012) On single image scale-up using sparse-representations. In: Boissonnat JD, Chenin A, Gout C, Lyche T, Mazure ML, Schumaker L (eds) Curves and surfaces. Springer, Berlin, pp 711–730

Zhang L, Zhang L, Mou X, Zhang D (2011) FSIM: a feature similarity index for image quality assessment. IEEE Trans Image Process 20(8):2378–2386. https://doi.org/10.1109/TIP.2011.2109730

Zhang J, Zhao C, Xiong R, Ma S, Zhao D (2012) Image super-resolution via dual-dictionary learning and sparse representation. In: 2012 IEEE international symposium on circuits and systems (ISCAS), pp 1688–1691

Zhang K, Zuo W, Chen Y, Meng D, Zhang L (2017a) Beyond a Gaussian denoiser: residual learning of deep CNN for image denoising. IEEE Trans Image Process 26(7):3142–3155. https://doi.org/10.1109/TIP.2017.2662206

Zhang K, Zuo W, Zhang L (2017b) Ffdnet: Toward a fast and flexible solution for CNN based image denoising. CoRR. arxiv: abs/1710.04026

Zhang Q, Yuan Q, Li J, Liu X, Shen H, Zhang L (2019) Hybrid noise removal in hyperspectral imagery with a spatial–spectral gradient network. IEEE Trans Geosci Remote Sens 57(10):7317–7329

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.